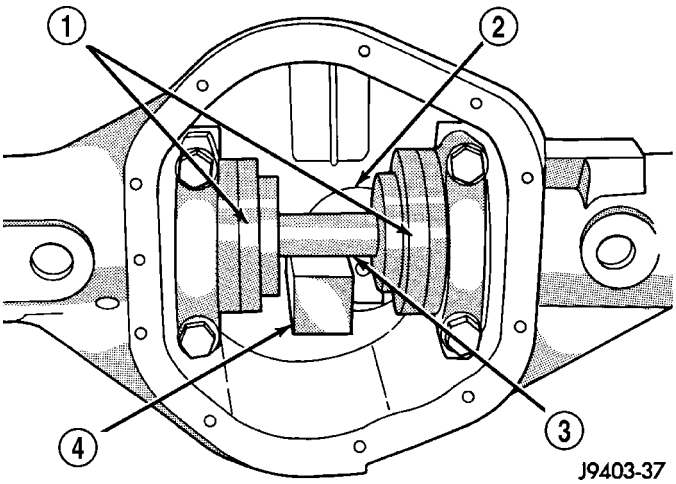


Fig. 12 GAUGE TOOLS IN HOUSING

1 - ARBOR DISC
2 - PINION BLOCK
3 - ARBOR
4 - PINION HEIGHT BLOCK

(6) Assemble Dial Indicator C-3339 into Scooter Block D-115-2 and secure set screw.

(7) Position Scooter Block/Dial Indicator flush on the pinion height block. Hold scooter block and zero the dial indicator.

(8) Slowly slide the scooter block across the pinion height block over to the arbor (Fig. 13). Move the scooter block till dial indicator crests the arbor, then record the highest reading.

(9) Select a shim/oil slinger equal to the dial indicator reading plus the pinion depth variance number etched in the face of the pinion (Fig. 8). For example, if the depth variance is –2, add +0.002 in. to the dial indicator reading.

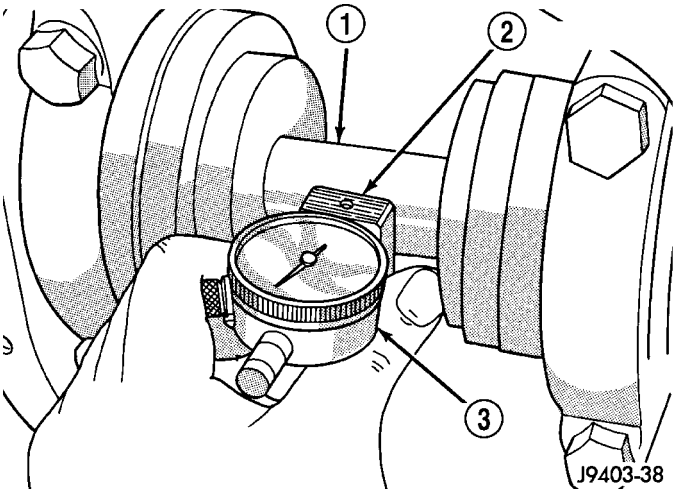


Fig. 13 PINION DEPTH MEASUREMENT

1 - ARBOR
2 - SCOOTER BLOCK
3 - DIAL INDICATOR

PRELOAD SHIM SELECTION

Differential side bearing preload and gear backlash is achieved by selective shims positioned between the differential side bearing cups and the housing. The proper shim thickness is determined using slip-fit Dummy Bearings D-345 in place of the differential side bearings and a Dial Indicator C-3339. Before proceeding with the differential bearing preload and gear backlash measurements, measure the pinion gear depth and prepare the pinion for installation. Establishing proper pinion gear depth is essential to establishing gear backlash and tooth contact patterns. After the overall shim thickness to take up differential side play is measured, the pinion is installed, and the gear backlash shim thickness is measured. The overall shim thickness is the total of the dial indicator reading and the preload specification added together. The gear backlash measurement determines the thickness of the shim used on the ring gear side of the differential case. Subtract the gear backlash shim thickness from the total overall shim thickness and select that amount for the pinion gear side of the differential (Fig. 14). Differential shim measurements are performed with spreader W-129-B removed.

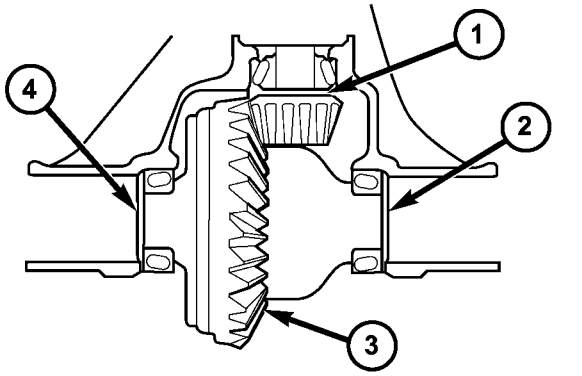


Fig. 14 SHIM LOCATION

1. PINION DEPTH SHIM
2. DIFFERENTIAL SHIM PINION GEAR SIDE
3. RING GEAR
4. DIFFERENTIAL SHIM RING GEAR SIDE

NOTE: It is difficult to salvage the differential side bearings during the removal procedure. Install replacement bearings if necessary.

(1) Remove differential side bearings from differential case.

(2) Remove factory installed shims from differential case.

(3) Install ring gear on differential case and tighten bolts to specification.

(4) Install dummy side bearings D-345 on differential case.

(5) Install differential case in the housing.

(6) Install the marked bearing caps in their correct positions and snug the bolts (Fig. 15).